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IoT-Based Light Intensity Monitoring and Smart Control System with Web Application

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1. Introduction

In recent years, the rapid advancement of the Internet of Things (IoT) has significantly transformed the way people interact with physical environments. IoT enables devices embedded with sensors, software, and communication technologies to collect and exchange data in real time. This technological evolution has led to the development of smart systems capable of monitoring, analyzing, and controlling various environmental parameters efficiently. Among these applications, smart lighting systems have gained considerable attention due to their potential to enhance energy efficiency, reduce operational costs, and improve user comfort.

Lighting plays a crucial role in daily human activities, especially in environments such as classrooms, offices, laboratories, and residential buildings. However, traditional lighting systems are often manually controlled, which can result in inefficient energy usage. Lights may remain turned on even when there is sufficient natural light or when the space is unoccupied. This not only wastes electrical energy but also increases operational costs and negatively impacts environmental sustainability. Therefore, there is a growing need for intelligent systems that can automatically monitor and regulate lighting conditions based on real-time data.

This project proposes the design and implementation of an **IoT-based light intensity monitoring and smart control system integrated with a web application**. The system will utilize a light sensor, such as an LDR or a digital light sensor, to measure ambient light intensity continuously. The sensed data will be processed by a microcontroller, specifically an ESP32, which is equipped with built-in Wi-Fi capabilities. The processed data will then be transmitted to a web-based application, where users can monitor light intensity levels in real time through an interactive interface.

In addition to monitoring, the system will incorporate a smart control mechanism that allows users to define threshold values for light intensity. When the measured light level falls below or exceeds the predefined threshold, the system can automatically trigger a response, such as turning on or off a light source (e.g., LED). This feature introduces automation into the system, reducing the need for manual intervention and ensuring optimal lighting conditions

at all times. Furthermore, the web application will provide features such as data visualization, status indication, and possibly historical data tracking, enabling users to make informed decisions regarding lighting usage.

The proposed system is designed to be cost-effective, scalable, and easy to implement, making it suitable for a wide range of applications. In educational institutions, for example, it can be used to manage classroom lighting efficiently. In office environments, it can contribute to energy-saving initiatives, while in residential settings, it can enhance convenience and comfort. Moreover, this project serves as a practical demonstration of integrating multiple engineering disciplines, including software development, electronics, and electrical control systems.

Overall, this project aims to provide a smart, efficient, and user-friendly solution for real-time light monitoring and control. By leveraging IoT technology, it contributes to the broader goal of developing sustainable and intelligent environments, while also enhancing the technical and practical skills of the students involved in its development.

2. Problem Statement

In many residential, educational, and commercial buildings, lighting systems are still controlled manually using traditional switches. While this method is simple, it often leads to significant **inefficiencies in energy usage**. Lights are frequently left turned ON even when there is sufficient natural daylight or when rooms are unoccupied. This results in unnecessary energy consumption, increased electricity costs, and reduced lifespan of lighting equipment. In environments such as classrooms and offices, where multiple users share the same space, it becomes even more difficult to ensure proper and efficient use of lighting systems.

Another major limitation of conventional lighting systems is the **lack of real-time monitoring and feedback**. Users do not have access to accurate information about the current light intensity in a given space. Without this information, it is difficult to determine whether the lighting conditions are optimal for comfort, productivity, or energy efficiency.

For example, insufficient lighting can negatively affect concentration and productivity, while excessive lighting can lead to discomfort and energy waste.

Furthermore, existing systems do not provide **automation or intelligent control mechanisms**. Lighting adjustments rely entirely on human intervention, which is not always reliable or timely. There is no system in place to automatically respond to changing environmental conditions, such as variations in natural light throughout the day. As a result, lighting systems fail to adapt dynamically to user needs and environmental changes.

In addition, there is a lack of **remote accessibility and centralized control**. Users cannot monitor or control lighting systems from a distance, which limits flexibility and convenience. In modern smart environments, users expect to access and control systems through digital platforms such as web or mobile applications, but traditional lighting systems do not support such capabilities.

The key problems identified can be summarized as follows:

- **Energy Inefficiency**
    - Lights remain ON unnecessarily
    - Poor utilization of natural daylight
    - Increased electricity consumption and costs
- **Lack of Real-Time Monitoring**
    - No system to measure or display light intensity
    - Users cannot assess lighting conditions accurately
    - No historical or analytical data available
- **Absence of Automation**
    - Manual operation of lighting systems
    - No automatic response to environmental changes
    - Dependence on human intervention
- **Limited Control and Accessibility**
    - No remote access to lighting systems
    - No centralized control interface
    - Inconvenience for users in managing lighting

- **User Comfort and Productivity Issues**
    - Inadequate lighting conditions
    - Negative impact on productivity and well-being

To overcome these challenges, there is a need for a **smart, automated, and intelligent lighting system** that can monitor light intensity in real time and provide efficient control mechanisms. This project aims to address these issues by developing an **IoT-based light intensity monitoring and smart control system with a web application**. The proposed system will enable continuous monitoring of environmental light conditions, provide real-time data to users, and implement automatic control based on predefined thresholds.

By integrating sensing, processing, communication, and user interface technologies, the system will significantly improve energy efficiency, enhance user convenience, and support the development of smart environments. Ultimately, this project seeks to bridge the gap between traditional lighting systems and modern intelligent solutions by introducing automation, real-time monitoring, and remote accessibility.

3. Objectives

3.1 General Objective

The general objective of this project is to design and implement an **IoT-based light intensity monitoring and smart control system integrated with a web application**, which is capable of measuring ambient light levels in real time, transmitting the data over a wireless network, and enabling users to monitor and control lighting conditions remotely. The system aims to enhance energy efficiency, improve user convenience, and demonstrate the practical integration of sensing, communication, and automation technologies in a smart environment.

3.2 Specific Objectives

The specific objectives of this project are:

- To measure ambient light intensity using a suitable sensor such as an LDR or BH1750
- To process and analyze sensor data using a microcontroller (ESP32)
- To transmit real-time light intensity data to a web application using Wi-Fi communication
- To design and develop a user-friendly web dashboard for monitoring light levels
- To implement threshold-based automatic control of lighting using an LED or similar output device
- To visualize light intensity data using charts and graphical representations
- To enable remote monitoring and control of the system through the web interface
- To improve energy efficiency by optimizing lighting usage based on real-time conditions

3.3 Scope of the Project

This project focuses on the design, development, and implementation of a **prototype IoT-based light intensity monitoring and control system**. The system will be limited to a controlled environment, such as a classroom or small indoor space, where light intensity can be measured and managed effectively.

The scope of the project includes the integration of a light sensor, microcontroller (ESP32), and a web-based application to enable real-time monitoring and control. The system will allow users to view light intensity data, set threshold values, and automatically control a lighting device (LED) based on predefined conditions. Communication between the hardware and the web application will be achieved using Wi-Fi.

However, the project will not cover large-scale industrial implementation or advanced features such as machine learning-based prediction, multi-room automation, or commercial deployment. The focus will remain on developing a functional, cost-effective, and scalable prototype that demonstrates the core concepts of IoT-based smart lighting systems.

4. Methodology

This project will adopt a **systematic design and implementation methodology** to ensure the successful development of an IoT-based light intensity monitoring and smart control system. The methodology is divided into four main phases: **system design, hardware development, software development, and system integration and testing**. Each phase will be carried out collaboratively by team members from different engineering disciplines to ensure a multidisciplinary approach.

4.1 System Design

The system design phase focuses on defining the overall architecture, components, and data flow of the proposed system. The system will be structured into three main functional units:

- **Sensor Unit:** Responsible for detecting ambient light intensity using a light sensor (LDR or BH1750).
- **Processing and Communication Unit:** Uses the ESP32 microcontroller to process sensor data and transmit it via Wi-Fi.
- **Application Unit:** A web-based interface that displays real-time data and allows user interaction and control.

A **block diagram** will be designed to visually represent the interaction between these components, including data flow from the sensor to the web application and control signals from the application back to the hardware. System requirements, including performance, accuracy, and response time, will also be defined at this stage.

4.2 Hardware Development

The hardware development phase involves the selection, design, and implementation of the physical components required for the system. The main hardware components include:

- Light sensor (LDR or BH1750)
- ESP32 microcontroller
- Resistors and supporting circuit elements

- LED for output indication
- Breadboard and connecting wires

The light sensor will be configured to measure ambient light intensity and generate corresponding analog or digital signals. These signals will be read by the ESP32 through its input pins (ADC for analog sensors or digital interface for BH1750). Proper circuit design will be ensured to maintain stable and accurate readings.

Additionally, the LED will be connected through a control circuit to act as an output device, demonstrating automatic lighting control. The hardware setup will be tested independently to verify sensor accuracy, signal stability, and proper functioning of the control mechanism.

4.3 Software Development

The software development phase consists of two major parts: **embedded system programming** and **web application development**.

Embedded Programming

The ESP32 microcontroller will be programmed using the Arduino IDE to perform the following tasks:

- Read light intensity data from the sensor
- Convert raw sensor values into meaningful units (lux or percentage)
- Process and compare the data with predefined threshold values
- Control the LED based on the threshold conditions
- Establish Wi-Fi connectivity
- Send real-time data to the web application using HTTP requests or APIs

Efficient and optimized code will be written to ensure real-time performance and reliable communication.

Web Application Development

A modern web application will be developed to provide an interactive interface for users. The system will use:

- **Next.js with TypeScript** for frontend and server-side rendering
- **Next.js API routes or Node.js** for backend functionality

The web application will include the following features:

- Real-time display of light intensity values
- Graphical visualization using charts
- Status indicators (e.g., Bright/Dark)
- User-defined threshold setting
- Control interface to send commands to the ESP32

The application will be designed with a focus on **usability, responsiveness, and performance**. Data communication between the ESP32 and the web application will be implemented using REST APIs.

4.4 System Integration and Testing

In this phase, all hardware and software components developed in the previous stages will be integrated to form a complete and fully functional IoT-based system. The goal of system integration is to ensure seamless interaction between the sensor unit, the ESP32 microcontroller, and the web application. Special attention will be given to achieving reliable communication, accurate data processing, and effective control mechanisms.

The integration process will establish a continuous data flow as follows:

Sensor → ESP32 → Web Application → ESP32 (Control Feedback) → LED

Initially, individual modules such as the sensor circuit, ESP32 communication module, and web application will be tested independently. After verifying their functionality, they will be gradually combined to ensure compatibility and proper synchronization. Integration testing

will focus on ensuring that data captured by the sensor is correctly processed by the ESP32 and accurately transmitted to the web application without delays or data loss.

The following testing activities will be carried out:

- **Real-Time Data Transmission Testing**
  - Verify that sensor data is transmitted continuously and reliably from the ESP32 to the web application
  - Check for delays, packet loss, or communication failures
- **Data Accuracy and Validation**
  - Ensure that the light intensity values displayed on the web application match the actual sensor readings
  - Compare readings under different lighting conditions to validate accuracy
- **Threshold-Based Control Testing**
  - Test the automatic control logic by setting different threshold values
  - Verify that the LED turns ON/OFF correctly based on the defined conditions
- **User Interaction and Command Validation**
  - Confirm that commands sent from the web application (e.g., threshold updates or manual control) are received and executed correctly by the ESP32
  - Ensure proper feedback is reflected in the system
- **System Stability and Reliability Testing**
  - Run the system continuously for extended periods to ensure stable performance
  - Identify and resolve issues such as system crashes, connection drops, or inconsistent behavior
- **Performance Testing Under Different Conditions**
  - Evaluate system performance in varying lighting environments (bright, dim, and dark conditions)
  - Measure system response time and accuracy under different scenarios

In addition to functional testing, **debugging and optimization** will be performed to improve system efficiency and responsiveness. Any errors or inconsistencies identified during testing

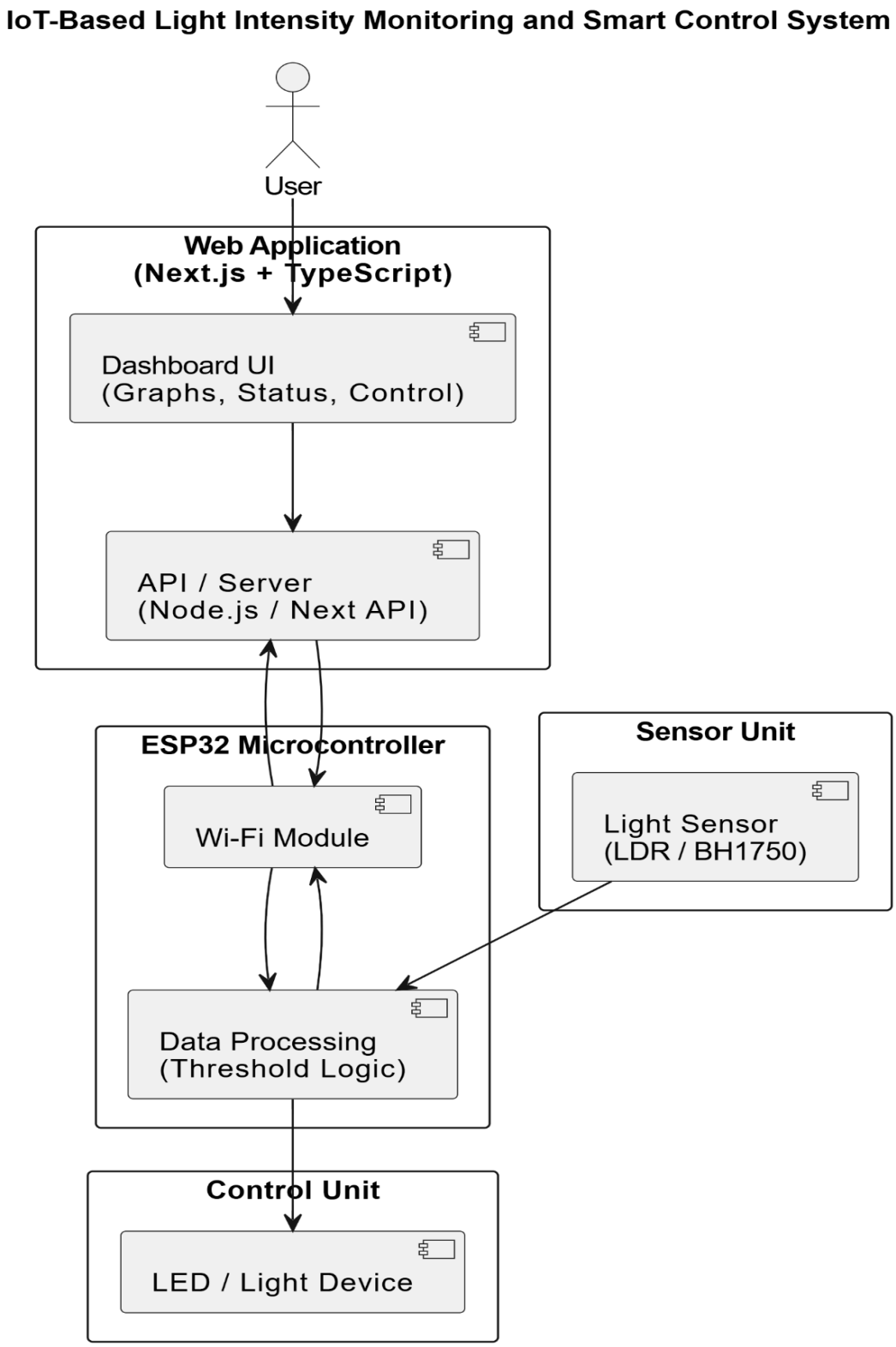
will be addressed through iterative refinement of both hardware and software components. This ensures that the final system is robust, reliable, and ready for demonstration.

4.5 Development Approach

The project will follow an **iterative and incremental development approach**, which allows the system to be developed in manageable stages while continuously improving its performance and functionality. This approach ensures flexibility, early detection of issues, and continuous feedback throughout the development process.

Key aspects of the development approach include:

- **Modular Development**
  - Each component (sensor, microcontroller, web application) will be developed separately
  - Simplifies debugging and improves system organization
- **Incremental Integration**
  - Components will be integrated gradually rather than all at once
  - Reduces complexity and allows early detection of integration issues
- **Continuous Testing and Feedback**
  - Each module will be tested immediately after development
  - Feedback from team members and advisors will be incorporated
- **Collaboration and Team Coordination**
  - Regular team meetings will be conducted (at least twice per week)
  - Tasks will be distributed based on each discipline (software, electronics, electrical)
  - Progress will be documented and shared with the project advisor
- **Version Control and Documentation**
  - Code and project files will be managed systematically
  - Proper documentation will be maintained for future reference and reporting
- **Time Management and Milestones**
  - The project will follow a defined timeline with clear milestones
  - Each phase will have specific deadlines



block diagram 4.1

5. Expected Outcomes

The successful completion of this project is expected to deliver a **fully functional, reliable, and scalable IoT-based smart lighting system** that demonstrates both technical innovation and practical applicability. The key outcomes of the project are outlined below:

- **Working Prototype System :** A complete and operational prototype capable of accurately measuring ambient light intensity using a sensor and processing the data through the ESP32 microcontroller.
- **Real-Time Web-Based Monitoring Application :** A user-friendly and responsive web application that displays real-time light intensity values, system status (Bright/Dark), and interactive data visualizations such as charts and graphs.
- **Smart Automatic Lighting Control :** An intelligent control mechanism that automatically turns ON or OFF a lighting device (LED) based on user-defined threshold values, ensuring optimal lighting conditions without manual intervention.
- **Remote Access and Control Capability :** The ability for users to monitor and control the lighting system remotely through the web application, providing flexibility, convenience, and enhanced user experience.
- **Improved Energy Efficiency :** Reduction in unnecessary energy consumption by optimizing lighting usage according to real-time environmental conditions, contributing to cost savings and environmental sustainability.
- **Reliable and Responsive System Performance :** A stable system capable of real-time data transmission, quick response to environmental changes, and consistent performance under different lighting conditions.
- **Scalability and Future Enhancement Potential :** A system architecture that can be easily expanded or upgraded to support additional features such as multi-room monitoring, integration with other smart devices, or implementation in larger smart building environments.
- **Practical Demonstration of IoT Concepts :** A clear demonstration of how IoT technologies can be applied to solve real-world problems by integrating sensors, microcontrollers, communication protocols, and web-based applications.

6. Required Materials and Tools

Hardware Components

- ESP32 microcontroller
- LDR or BH1750 light sensor
- Resistors
- LED
- Breadboard and jumper wires

Software Tools

- Arduino IDE
- Web development tools (VS Code)
- Browser for testing

Materials Cost Breakdown

Component	**Quantity**	**Unit Price (ETB)**	**Total Price (ETB)**
ESP32 Microcontroller	1	1500	1500
Light Sensor (LDR / BH1750)	1	400	400
LED	2	200	200
Resistors (assorted pack)	1 pack	300	300
Breadboard	1	1000	1000
Jumper Wires	1 set	250	250
Total			**3650 ETB**

7. Team Roles and Responsibilities

Software Engineering Students

- Develop web application (frontend and backend)
- Implement data visualization (graphs, UI)
- Handle communication between ESP32 and web app

Electronics & Communication Students

- Sensor integration and circuit design
- ESP32 programming
- Wi-Fi communication setup

Electrical Power & Control Students

- Power supply design
- LED control system
- System safety and reliability

8. Timeline

Task	**Duration**
Idea and proposal	1 week
System design	1 week
Hardware setup	2 weeks
Software development	3 weeks
Integration and testing	2 weeks
Final report and presentation	1 week

9. Conclusion

This project presents a practical and efficient IoT-based solution for real-time monitoring and smart control of light intensity. By integrating sensors, the ESP32 microcontroller, and a web application, the system enables automated operation, remote accessibility, and improved energy management. The proposed solution is cost-effective, scalable, and easy to implement, making it suitable for educational and smart environment applications.

In addition, the system demonstrates how modern IoT technologies can be applied to solve real-world problems related to energy efficiency and automation. It provides a flexible platform that can be further enhanced with advanced features such as multi-device integration and cloud-based monitoring. Furthermore, the project strengthens multidisciplinary engineering skills by combining embedded systems, communication technologies, and software development, preparing students for real-world engineering challenges.